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Audit Feedback, Decisions, and Explanations in LLM Trading Agents

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Abstract: We ask whether audit feedback changes what large language model (LLM) trading agents do or only what they say. Motivated by an earlier exploratory phase (Supplementary Section S3), this preregistered synthetic-market experiment has a switching fundamental-return driver. It compares five audit conditions. A sign-flip probe reverses a feature's sign and re-asks the same prompt; an identical-prompt null re-ask measures response variability. Precision measures agreement between cited features and features whose reversal changes the action. The strategic-compliance index (SCI) compares citation change with sign-flip-dependence change. Holm adjustment accounts for multiple comparisons. DeepSeek's primary contrasts were evidence-minus-naive precision -0.014 (Holm $p = 0.368$), true-driver sign-flip dependence -0.062 (Holm $p = 0.014$), label SCI -0.039 (Holm $p = 0.273$), and placebo SCI -0.134 (Holm $p = 4.18 \times 10^{-5}$). Relative to naive, the placebo package increased opposite-driver sign-flip dependence more than citation under the registered SCI. Evidence reduced true-driver sign-flip dependence in Luna ($\Delta = -0.091$, Holm $p = 0.038$). Both contrasts estimate audit packages, not isolated content effects. Instrument checks passed the pooled flip-rate gate only; news-regime discrimination reversed in DeepSeek and was near zero in Haiku. Auditor agreement does not establish process alignment or identify internal reasoning or deceptive intent.

Keywords: large language models; explanation faithfulness; counterfactual attribution; accountability; auditing; agent-based simulation; intelligent agents; multi-agent systems; autonomous agents; human–AI oversight interfaces

1. Introduction

Large language model (LLM) agents increasingly act as autonomous software agents in networked services and online markets, including API-connected trading and e-commerce agents and multi-agent platforms. In such intelligent-agent systems, oversight can occur remotely through logged explanations and automated auditor services. These services form an intelligent human–computer interface between agents and human or automated supervisors: explanations communicate reasons, while auditor feedback messages can steer subsequent behaviour. This connects explanation evaluation to the design of oversight channels for Internet applications and smart systems. Our controlled simulation examines this feedback relationship; it does not evaluate a network deployment, communication protocol, or networking performance.

An auditor may accept an agent's explanation without knowing whether the stated reasons influenced its action. This problem arises when large language models (LLMs) select actions in decision-support and agentic systems. Users may trust an output because of its explanation, and auditors or operators may use that explanation when deciding whether to intervene. In the European Union, Article 13 of Regulation (EU) 2024/1689, the Artificial Intelligence Act, requires high-risk AI systems to provide sufficient transparency for deployers to interpret their outputs and use them

appropriately [1]. Explanations therefore need evidential value. Our synthetic trading task is not a legal compliance assessment or a claim that every trading application falls within that provision.

An explanation can agree with an evaluator yet say little about the action it accompanies. Generated rationales can omit influential factors [2]. Research on self-generated counterfactual explanations and decision explanations raises related concerns [3,4]. The evaluator may also supply the expected answer. Agreement can then reflect adoption of an authoritative label. We test whether the associated decision changes as well.

Our preregistered confirmatory experiment uses a controlled market. The fundamental component of returns depends on news or momentum and switches repeatedly between them. Agents observe features, trade, and cite their reasons. We measure two kinds of agreement. *Environmental fidelity* means that an agent cites the feature currently driving fundamental returns. *Restricted process alignment* means that cited features agree with features whose sign reversal changes the agent's action. The latter uses a *counterfactual sign-flip probe*: we reverse one feature's sign, keep the rest of the prompt fixed, and re-ask for an action. Environmental fidelity concerns the environment; restricted process alignment concerns the agent's response to this intervention. Neither measure provides direct access to internal reasoning.

An earlier exploratory phase with a different design found that agents adopted auditor-disclosed labels even when false. Its corrective-audit environmental-fidelity gain became inconclusive after decisions exposed to disclosed labels were excluded. We call this descriptive pattern auditor echo. It motivated us to measure decisions as well as explanations. Preliminary data are never pooled with confirmation and provide no confirmatory evidence; Supplementary Section S3 gives the design, results, and limitations.

We therefore distinguish citation adoption from decision sensitivity under institutional feedback. Following the exploratory motivation (Supplementary Section S3), we test whether auditor echo extends to decisions. In DeepSeek, the placebo audit package increased opposite-driver sign-flip dependence more than citation relative to naive under the registered SCI. The evidence audit package also reduced true-driver sign-flip sensitivity relative to naive in the primary model and one replication. Neither comparison isolates feedback content from auditing and fines. We report null results, contrary effects, instrument weaknesses, and model differences alongside these findings.

1.1. Background and related work

Explainable AI and explanation faithfulness.

Research distinguishes models that people can interpret from post hoc explanations, which are generated after a prediction [5]. Other work examines what interpretability means [6] and surveys explanation methods [7–9]. Social and pragmatic accounts ask what explanations are for and how to evaluate them [10,11]. Local surrogates such as LIME approximate a prediction near one input [12]. Additive attribution methods such as SHAP divide a prediction among input features under particular assumptions [13]. Generated rationales require separate evaluation. Chain-of-thought explanations can omit experimentally introduced influences [2], and a model's own counterfactual account need not locate its decision boundary [3]. Jacovi and Goldberg distinguish faithfulness, or accuracy about the basis for a decision, from plausibility to a reader [14]. We use this distinction to report environmental agreement and behavioral-probe agreement separately.

Counterfactual measurement of process alignment.

A model can propose a counterfactual explanation without anyone changing its input. Our counterfactual experiment instead holds the decision context fixed and re-queries the action after reversing one feature's sign. The resulting change belongs to the model–prompt–provider system under that intervention. It does not fully decompose the causes of the decision. Correlated features, interactions, unprobed inputs, response variability, and altered input plausibility can still affect the

result. Our precision and recall therefore cover only the two probed features. The explanation–behavior gap is a behavioral index, not a report of hidden computation.

LLM agents in markets and strategic settings.

LLM market simulations study financial behavior [15], and market games examine bounded strategic reasoning [16]. Stylized facts are recurring statistical patterns in empirical returns [17]. They can validate realistic finance simulations, whereas our market prioritizes controlled fundamental-return generation. Generative agents also operate in interactive social environments [18]. Related work addresses double-auction collusion [19], strategic communication and information leakage [20], and adversarial coordination [21]. We examine how institutional messages relate to declared reasons and measured decision sensitivity. We do not aim for trading realism or infer strategic intent.

Auditing, transparency, and accountability.

Work on the limits of transparency [22], legal debates about explanation [23], and ethical frameworks for AI [24] warns against equating disclosure with understanding. Internal auditing frameworks [25] and recursive audit architectures [26] provide institutional context. Cheap-talk theory [27] and narrative economics [28] help frame why messages may matter, although our study does not test those theories directly. The confirmatory phase compares feedback that names a driver with feedback that reports observational evidence. Neither kind is assumed to equal independent verification by a human institution.

2. Materials and Methods

2.1. Design, agents, and feedback

The preregistered confirmatory experiment crosses five conditions with a primary DeepSeek cohort of twenty paired market seeds and independent Haiku and Luna replications of ten seeds each. Development seeds are excluded. Six heterogeneous agents trade for forty ticks per run. Agents share exogenous market draws across conditions, while their actions change subsequent prices, portfolios, and observed return histories. Seeds, rather than decisions or agents, are the replication units. The primary analysis uses the latter half of each run, retaining regime switches and stale feedback. Supplementary Section S1 supplies the complete design, generation equations, seeds, windows, prompts, and analysis rules.

The synthetic asset's fundamental return component switches between generated news and momentum signals; order imbalance and noise also affect returns. Agents observe the current features and four completed feature–return rows. They submit strict JSON buy/sell actions, integer sizes, and up to two reason labels. Heterogeneous mandates and starting inventories do not create independent decision makers. Borrowing and short selling are prohibited, and re-asks never execute trades.

All conditions require explanations and share the same institutional framing (Supplementary Table S1). Naive receives no audit. Punitive receives an environmental verdict and a total-wealth fine after an audited failure. Label additionally discloses the true driver; placebo discloses the opposite driver despite a truth-based verdict. Evidence reports recent feature–return correlations without naming a driver. Feedback first appears in the next prompt and persists until another audit, including across switches. The evidence package summarizes a longer history than the common prompt and includes endogenous returns, so its contrast is not an isolated test of numerical presentation. Placebo likewise tests a conflicting feedback package rather than label falsity alone.

2.2. Measurement and inference

An independent, prespecified mask selects half of decisions for counterfactual measurement. Each valid selected response is re-asked twice, reversing only news or momentum while retaining the

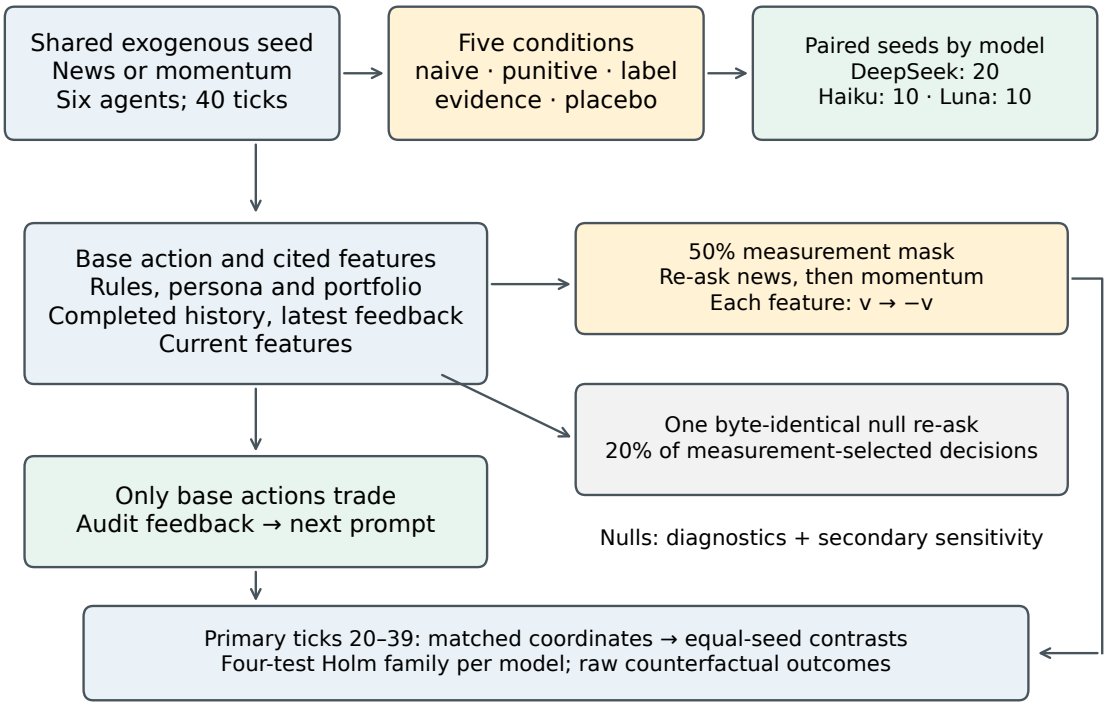


Figure 1. Confirmatory phase design. Conditions share exogenous innovations and prespecified sampling streams. Only base decisions trade; sign-flip and null re-asks retain the base context. The displayed null probability is the amended design; the exact realized null-mask recipe deviated as disclosed in Supplementary Section S2. Model sample sizes are inferred from the included manifest.

exact remaining context. An identical-prompt null re-ask on a subset monitors response variability. Only the base decision trades. Environmental fidelity records citation of the current fundamental driver. True-driver dependence records an action change after reversing that driver's feature. Process precision is the share of cited probe features whose reversal changes the action; recall reverses that denominator. These scores cover only the probed features, remain undefined at empty denominators, and measure agreement with this intervention rather than ground-truth internal reasoning.

The strategic-compliance index (SCI) subtracts treatment-minus-naive sign-flip-dependence change from the corresponding citation change on matched coordinates. Primary label SCI selects the current true driver, and placebo SCI its opposite, regardless of prior audit exposure. Secondary exposure SCI instead uses the actually disclosed, possibly stale label. Neither sign identifies intent or deception. Within-seed scores average defined decision-level values, use common eligible coordinates across paired conditions, and give each seed equal weight. Treatment-dependent eligibility can affect the estimand. Parse/transport failures and structurally undefined scores remain distinct; no actions are fabricated or missing scores silently filled.

The four primary contrasts test evidence-minus-naive precision and dependence, label SCI, and placebo SCI. All predicted directions were positive. Two-sided paired seed-level tests use Holm correction within each model; models are not pooled and replications cannot rescue a failed primary prediction. Bootstrap intervals are descriptive, non-simultaneous, and do not override those tests. Sensitivities and secondary/exploratory analyses do not replace the frozen family. Supplementary Section S1 gives formulas, hypotheses, exclusions, missingness rules, uncertainty procedures, and prospective sensitivity limits.

DeepSeek used temperature zero; the replication routes used provider-default sampling because temperature control was not advertised. JSON enforcement, disabled reasoning/thinking, pinned serving providers, identical-payload transport retries, fixed masks, and freeze guards constrained

Table 1. The four registered DeepSeek contrasts, with a compact replication summary. P1: evidence minus naive process precision; P2: evidence minus naive true-driver dependence; P3: label SCI; P4: placebo SCI. All predicted signs were positive. Intervals are descriptive 95% paired-seed bootstrap intervals, not simultaneous intervals; Holm correction is within each model. Replication summary: Haiku had no corrected primary rejection; Luna had negative evidence precision and dependence, with inconclusive SCI tests. Neither replication supports predicted positive effects. Replication estimates and inferential details are in Supplementary Table S2.

Model	Test	n	Δ	95% CI	Raw p	Holm p	d_z
DeepSeek	P1	20	-0.0137	[-0.0431, 0.0140]	0.3683	0.3683	-0.2061
DeepSeek	P2	20	-0.0624	[-0.0981, -0.0239]	0.0047	0.0140	-0.7164
DeepSeek	P3	20	-0.0389	[-0.0881, 0.0068]	0.1364	0.2727	-0.3478
DeepSeek	P4	20	-0.1341	[-0.1772, -0.0916]	1.04×10^{-5}	4.18×10^{-5}	-1.3257

collection. Services are not immutable checkpoints. Amendments followed development calibration with descriptive label outcomes visible to the PI/coordinator; a later amendment changed only costs/operations. Deviations in null-mask realization and a selected invalid-base null call affect protocol fidelity and noise monitoring. Raw primary outcomes were not repaired or redefined. Supplementary Sections S1–S2 disclose these details and execution incidents.

3. Results

All assigned confirmatory runs completed without replaced seeds, and every primary contrast retained all seed pairs. Every DeepSeek primary estimate opposed its predicted positive direction. Evidence precision and label SCI were negative and inconclusive. Evidence dependence and placebo SCI rejected after Holm correction in the negative direction. Table 1 reports the registered estimates and uncertainty; inconclusive findings do not establish equivalence.

Figure 2 separates citation change from dependence change on the same coordinates and seed weights. In DeepSeek, the placebo package increased opposite-driver sign-flip dependence more than citation relative to naive. This does not establish explanation incompleteness or deliberate concealment. Correct labels provided no evidence of disproportionate citation improvement, while an inconclusive SCI cannot exclude all explanation-only changes.

Neither replication supported a predicted positive primary effect. Haiku had no corrected primary rejection. Luna showed negative evidence effects on precision and dependence, while its SCI tests remained inconclusive. DeepSeek's negative placebo gap therefore does not generalize across both replications. Detailed estimates appear in Supplementary Table S2; the descriptive forest plot does not replace the registered tests.

The probe passed only the pooled flip-rate gate: news-regime discrimination reversed in DeepSeek and was near zero in Haiku. Momentum discrimination was positive in both, and Luna was positive in both regimes. We interpret affected strata as sign-flip sensitivity rather than correct driver use. All strata remain in the frozen tests. Raw flips combine intervention sensitivity with re-ask variability; noise differs by condition and model. Undefined precision/recall and common-support restrictions limit unconditional interpretation. Supplementary Tables S4, S9, and S10 separate validity, coverage, and null monitoring.

Registered sensitivities retain the primary estimate signs, although raw significance decisions can change in the wider window. Secondary outcomes show label adoption without a consistent evidence precision benefit; the relative DeepSeek evidence-over-punitive precision advantage does not imply improvement over naive or punishment equivalence. Review-requested regime and paired-noise analyses preserve the main DeepSeek directions but reveal heterogeneous Luna SCI and noise sensitivity of label SCI. They remain exploratory and do not replace any registered verdict. Full results and qualifications are in Supplementary Section S2 and Tables S3, S5–S8, and S11.

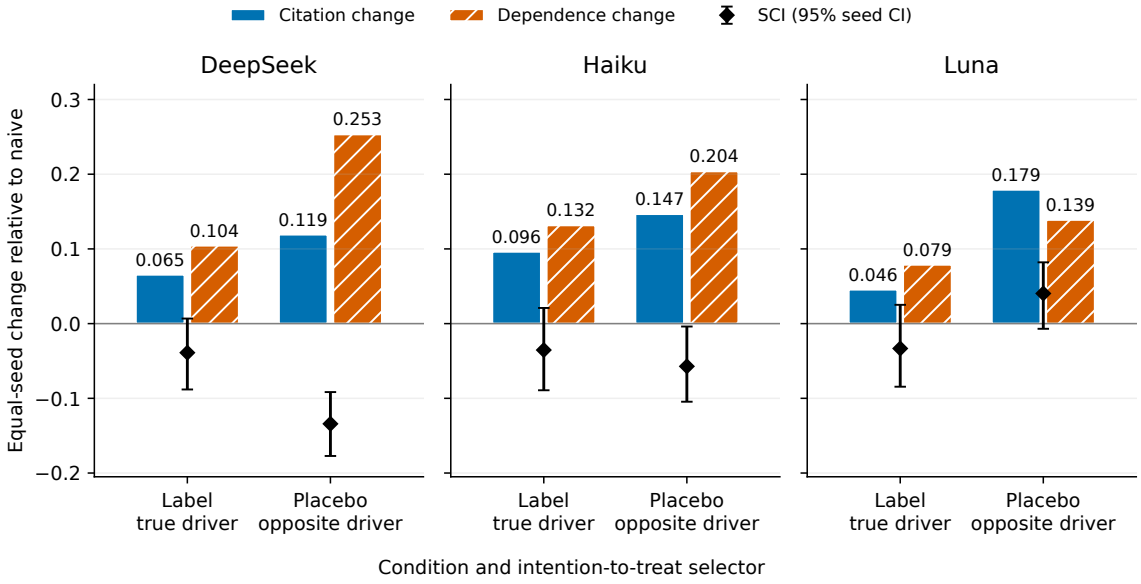


Figure 2. SCI decomposition on matched primary-window coordinates, with equal seed weights. Bars show citation and dependence changes; diamonds show their difference (SCI) and its descriptive 95% seed-bootstrap interval. Label selects the current true driver and placebo its opposite, irrespective of audit exposure. Component bars are descriptive and are not additional primary tests.

4. Discussion

4.1. What an auditor's agreement establishes

Agreement with an auditor does not show what controlled an action. Environmental fidelity alone cannot establish that relationship. Under the registered sign-reversal instrument, the label and placebo audit packages change measured sign-flip sensitivity as well as citations relative to naive. Correct-label SCI provides no evidence of disproportionate citation improvement in any model. Inconclusive tests cannot exclude all explanation-only changes. The principal placebo-package result is narrower. DeepSeek's negative SCI indicates that the increase in opposite-driver sign-flip dependence exceeded the increase in opposite-driver citation under the registered SCI. The gap is defined relative to naive. It does not mean that explanations never mention the false feature.

Oversight should assess environmental agreement and behavioral alignment separately. An auditor can change the system it evaluates, so its message is an intervention as well as a measurement. Audits should test message accuracy, including stale and incorrect feedback, alongside whether the agent repeats the correct answer. The Article 13 motivation is methodological: information supplied for interpretation should have an evidential basis. This experiment does not establish a legal sufficiency test or a complete operational audit standard.

For networked auditing services, the findings motivate evaluating feedback channels as interventions as well as records of oversight. An apparently authoritative message may steer an autonomous agent's decisions even when its label is incorrect, so matching an auditor's vocabulary is insufficient evidence that an oversight interface supports reliable supervision. Evaluations of multi-agent systems should therefore distinguish explanation agreement from decision sensitivity and test stale or erroneous feedback alongside accurate messages. These are design implications for intelligent agents and human–AI oversight interfaces, subject to the instrument and model limitations reported here; the experiment establishes neither deployment safety nor the effectiveness of a particular network architecture.

4.2. False-authority risk

The placebo audit package can change decisions and citations, but this experiment cannot isolate why. The placebo intervention combines an environmental verdict with a conflicting authoritative driver label. Its effects could reflect message salience, ambiguity resolution, or reliance on the disclosed feature. We cannot isolate those mechanisms. Under the registered SCI, the placebo package increased opposite-driver sign-flip dependence more than citation relative to naive in DeepSeek. These two response rates are not calibrated measurements of a common latent quantity. Salience, response variability, fixed-history inconsistency or citation-format constraints could contribute. The result motivates testing responses to erroneous authority. It does not establish under-explanation, concealment or a mechanism attributable to label falsity alone.

The model differences constrain this result. Haiku's placebo estimate is negative but fails its corrected primary test. Luna's is positive and inconclusive. The DeepSeek result therefore does not establish how LLMs generally respond to audits. Persisted labels and the deliberate verdict/label inconsistency also differ from many practical audits. These features make false authority testable here while limiting direct deployment generalization.

4.3. Why numerical evidence may hurt

The evidence audit package did not yield the predicted gains in registered process measures. The registered evidence prediction failed. The evidence condition had lower true-driver dependence than naive in DeepSeek and Luna. The registered comparison does not isolate the effect of its correlation table from audits and fines. Luna precision was also lower, while Haiku remained inconclusive. Instrument checks passed on the pooled flip-rate gate only. Precision/recall measure agreement with the probe rather than ground-truth process accuracy. Exploratory evidence-minus-punitive dependence estimates remain negative, but their intervals are descriptive and the history/content difference is not isolated. Several explanations remain compatible with the design. Short-window correlations can be noisy, the evidence window can span a regime switch, and observed returns contain endogenous order imbalance. A table over every feature may distribute attention toward distractors or preserve an outdated association. Its longer summarized history also changes information availability rather than presentation alone. Future factorial tests could examine these interpretations; they are not established mediators.

Lower dependence does not automatically imply poorer economic performance. The true feature governs only the fundamental return component, and appropriate trading can also depend on portfolio constraints or impact. The observed decreases concern the registered sign-flip-sensitivity/precision targets in this setting. Reversed or absent discrimination precludes describing every decrease as poorer driver use. Evidence has a relative precision advantage over punitive in DeepSeek, but it does not show an advantage over naive.

4.4. Limitations

The market is synthetic and deliberately simple. News/momentum are generated variables, and a known switch schedule governs the fundamental component. Forced buy/sell choices can elicit decisions when evidence is weak. Short horizons and an oracle environmental auditor that knows the true driver limit resemblance to real markets. Treatment changes market paths through agent orders. A paired seed controls exogenous draws but does not isolate a direct linguistic mechanism. We tested only three deployed models and one prompt family, without immutable checkpoints or broad prompt/order counterbalancing.

The instrument probes only news and momentum, holds history and correlated distractors fixed, and records action rather than size. Sign reversal avoids the original adversarial extremes but can still produce an unusual joint feature/context combination. The probe can miss interactions, asymmetric thresholds, correlated predictors, and altered confidence without an action change. Sensitivity does

not show that a feature was necessary or capture complete process faithfulness. Passing the pooled flip-rate gate does not remedy the news-regime discrimination reversal in DeepSeek or near-zero discrimination in Haiku. Review-requested strata preserve the negative DeepSeek P2/P4 directions in both regimes but reveal opposing Luna P4 signs. These exploratory results narrow the interpretation rather than validate the instrument. We therefore cannot interpret every flip as correct use of the environmental driver.

Null re-asks reveal response variability, including with DeepSeek temperature zero. Haiku and Luna used provider-default sampling without temperature control. Rate subtraction is a secondary additive sensitivity. It does not correct individual decisions or statistically separate signal from noise through deconvolution. The new paired sensitivity resamples seed-level rates and contrasts together. All-tick cell subtraction cannot eliminate time/regime heterogeneity or estimate a pure feature effect. Model-by-condition noise differences and the disclosed null-mask/invalid-base deviations complicate that interpretation. Provider-default replication sampling also prevents attributing cross-model differences to model identity alone. Precision and recall have treatment-dependent undefined denominators. Common support reduces comparability problems without making the resulting estimand unconditional. The gap can reflect either component. Its sign does not identify internal reasoning or intent, and it cannot establish deception.

Development calibration and noise checks reused dev seeds, and the PI/coordinator saw descriptive label outcomes before freezing. The amendments preserve the original plan and hypotheses transparently. They do not erase that exposure or make instrument choice independent of the development context. The second amendment occurred during collection and was restricted to costs and operations. Supplementary Section S3 gives additional preliminary-phase limitations. Future preregistered work should validate instruments across regimes and prompt families before testing treatment effects.

Fixed personas and inventory differences do not make six independent decision makers. Within-run dependence remains. Seed counts limit sensitivity to small effects, and the two replications have fewer seeds than the primary model cohort. An inconclusive contrast is not equivalence. A positive or negative SCI is an operational behavioral comparison, not evidence for a human motivational category.

5. Conclusions

The confirmatory experiment shows why explanation change and decision change must be measured separately. Beyond the exploratory motivation (Supplementary Section S3), all four primary DeepSeek estimates oppose their predicted positive directions. The evidence-dependence effect and placebo SCI reject after correction, in directions opposite to prediction. Evidence precision and label SCI remain inconclusive.

The evidence condition also reduces true-driver sign-flip dependence relative to naive in Luna, while Haiku has no corrected primary rejection. This estimates an audit package, not isolated numerical content. In DeepSeek's placebo condition, the increase in opposite-driver sign-flip dependence exceeded the increase in opposite-driver citation under the registered SCI. This is also a package contrast. The negative gap does not generalize across both replications or establish explanation incompleteness or deceptive intent. Exploratory regime and paired noise analyses preserve the main DeepSeek directions but reveal heterogeneous Luna SCI and sensitivity of label SCI to noise adjustment. No registered conclusion is replaced. Auditor agreement is insufficient evidence of process alignment, and the evidence package is not automatically beneficial. Evaluation should distinguish environmental agreement, intervention-defined dependence, coverage, noise, and incorrect-authority responses while respecting the limits of the probe.

Supplementary Materials: The accompanying file supplementary_v15.pdf contains detailed methods, registered and exploratory results, preliminary-phase materials, prompts, reproducibility and availability records, and coverage/null monitoring (Sections S1–S5; Tables S1–S11). Its source is supplementary_v15.tex.

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Use of Artificial Intelligence: The project used an AI-agent workflow under PI and coordinator supervision for software implementation, development checks, experiment orchestration, analysis scripting and reporting, internal adversarial review, and manuscript drafting. AI coding and writing assistants were used under author supervision. This submission package was prepared by a dispatched coding assistant using the existing frozen study materials, offline document generation, and LaTeX compilation. Experimental LLMs were the experimental subjects accessed through APIs and produced the trading responses analyzed in the study; assistant-generated reviews and prose are not additional experimental observations. The artifacts identify execution and analysis provenance and disclose protocol deviations. The named human authors take full responsibility for scientific interpretation, verification, the manuscript, and submission.

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